

ECON 351 · Focused Practice Set

Set 8 of 10

Chapter 9

Risk Attitudes

4 questions in this practice set

Questions: Q18, Q19, Q20, Q21

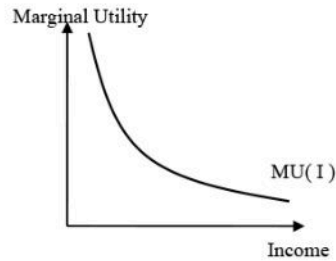
Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 4 (Practice Q18)

Risk Attitudes #1 · HW2 Ex 2

Exercise 2 - Attitudes towards risk and fair bets

The following graph represents Mike's **marginal utility** of income (not the utility but the marginal utility):



We know that Mike's income I is greater than \$20, but we do not know his exact income. Alejandro offers Mike the following bet: Alejandro will roll a 6 sided dice (all numbers equally likely). If it is 1 or 2, Mike pays Alejandro \$20. Otherwise, if it is 3, 4, 5 or 6, Alejandro pays Mike \$10.

a) Is this a fair bet?

b) Will Mike accept or reject the bet? Does your answer depend on Mike's initial wealth? (assuming he can always pay the \$20 if he loses)

Question 2 of 4 (Practice Q19)

Risk Attitudes #2 · Sample Q6

- a) Compute the expected value of each investment option.
- b) Compute the expected utility of each investment option. Find the best (and worse) investment option for Jim.

Question 3:

Pedro Carroll has the following utility from income: $u(I) = \sqrt{I}$. Currently, he has an income $I = \$169$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Seahawks to win the championship game. If he bets all his money and the Seahawks win, then he ends up with an income $I = \$400$. If he bets all his money and the Seahawks lose, then he ends up with nothing, $I = \$0$. If he does not bet, he keeps his \$169 for sure.

The Los Angeles Seahawks will win the championship game with probability P , and will lose the game with probability $1-P$.

Compute the minimum winning probability P such that Pedro would be willing to bet all his money on the Seahawks.

- (A) $P = 50\%$
- (B) $P = 65\%$
- (C) $P = 70\%$
- (D) $P = 55\%$
- (E) $P = 60\%$

Question 4:

Cam has the following utility from income: $u(I) = \sqrt{I}$. Currently he has an income $I = \$36$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Panthers to win the championship game. If he bets all his money and

the Panthers win, then he ends up with an income I_{Win} . If he bets all his money and the Panthers lose, then he ends up with nothing, $I_{Lose} = \$0$. If he does not bet, he keeps his \$36 for sure.

The Los Angeles Panthers will win the championship game with probability 75%, and will lose the game with probability 25%.

Compute the minimum winning prize I_{Win} such that Cam would be willing to bet all his money on the Panthers.

- (A) $I_{Win} = \$36$
- (B) $I_{Win} = \$48$
- (C) $I_{Win} = \$56$
- (D) $I_{Win} = \$64$
- (E) $I_{Win} = \$72$

**Question 6:**

Anna's utility of income function is $U(I) = I$, while Elsa's utility of income function is $U(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

Question 3 of 4 (Practice Q20)

Risk Attitudes #3 · Sample Q8

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

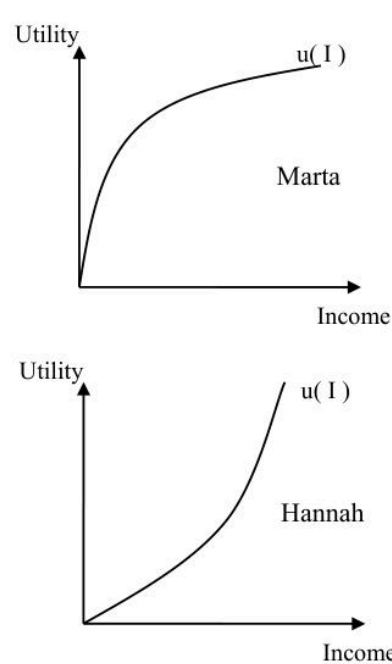
Question 7:

Anna's marginal utility of income function is $MU(I) = I$, while Elsa's marginal utility of income function is $MU(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 8:

The following graphs represent the utility function of consumers Marta and Hannah:

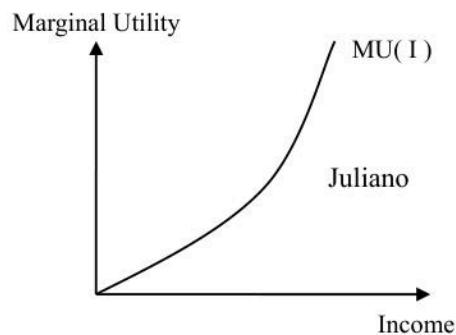
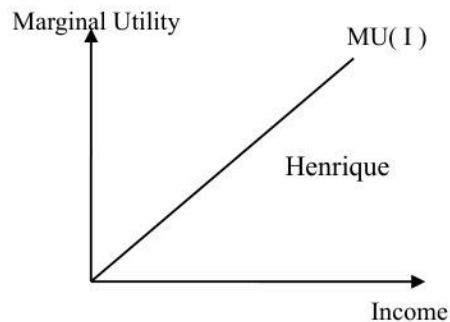


Using the information from the graphs,

- (A) Both Marta and Hannah are risk loving
- (B) Both Marta and Hannah are risk averse
- (C) Marta is risk averse; Hannah is risk loving
- (D) Marta is risk loving; Hannah is risk averse
- (E) Not enough information from the graphs

Question 9:

The following graphs represent the marginal utility of income functions of consumers Henrique and Juliano.



Using the information from the graphs, Henrique is _____ and Juliano is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 10:

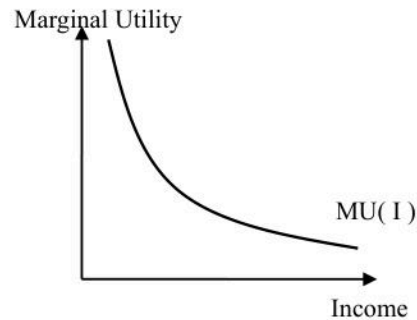
Mike has a constant marginal utility of income. We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough

- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 11:

The following graph represents Mike's marginal utility of income:



We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough
- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 4 of 4 (Practice Q21)

Risk Attitudes #4 · Mock Q28

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.